



Sketching linear graphs

What to do

- What is a linear function?
- By drawing up a table of values, plot the following linear functions.
 - $f(x) = 2x + 1$
 - $f(x) = 3x - 1$
 - $f(x) = 2 - 3x$
 - $f(x) = 4 - x$
- a Plot the graph for this table of values.

x	-2	-1	0	1	2
y	12	9	6	3	0

- b Find the equation that connects x and y .
- Find the gradient and y -intercept for each of the following lines.
 - $y = 2x - 1$
 - $y = \frac{3}{4}x - 11$
 - $y = 4 - 2x$
 - $2y = x + 5$
- Determine the x - and y -intercepts for each of the following equations and sketch their corresponding graph.
 - $y + x = 4$
 - $y = 5 - x$
 - $3y + 2x = 12$
 - $3x - 4y = -12$
- a On the same axes, graph $y = -2x - 1$, $y = -2x + 1$ and $y = -2x + 3$.
b What is the name given to these lines?
- MC** Which of the following equations does not have a y -intercept of -2 ?
 A $4x = y + 2$ B $2x + y = -2$ C $y = x - 2$ D $4x - y = -2$
- Graph each of the following functions for the given domain.
 - $f(x) = 2x + 1$, $-2 \leq x \leq 3$
 - $f(x) = -x + 5$, $-3 \leq x \leq 2$
 - $f(x) = -4x$, $-2 < x < 2$
- a Find the value of b given that the point $(0, 2)$ lies on the line $y = 5x - b$.
b What is the x -intercept?